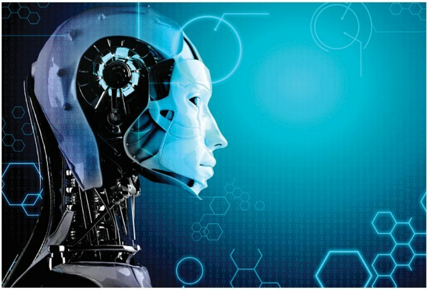
Compiled by:
Jagmeet Singh

Microsoft's Project Brainwave offers real-time AI



Expanding its footprint in the artificial intelligence (AI) world, Microsoft has unveiled a new deep learning acceleration platform called Project Brainwave. The new project uses a set of field-programmable gate arrays (FPGA) deployed in the Azure cloud to enable a real-time AI experience at a faster pace.

The system under Microsoft's Project Brainwave is built on three main layers: a high-performance distributed system architecture, a DNN engine synthesised on FPGAs, and a compiler and runtime for low-friction deployments of trained models. The extensive work on FPGAs by the Redmond giant enables high performance through Project Brainwave. Additionally, the system architecture assures low latency and high throughput.

One of the biggest advantages of the new Microsoft project is the speed. FPGAs on the system are attached directly with the network fabric to ensure the highest possible speed. The high throughput design makes it easier to create deep learning applications that can run in real-time.

"Our system, designed for real-time AI, can handle complex, memory-intensive models such as LSTMs, without using batching to juice throughput," Microsoft's distinguished engineer Doug Burger wrote in a blog post.

It is worth noting that Project Brainwave is quite similar to Google's Tensor Processing Unit. However, Microsoft's hardware supports all the major deep learning systems. There is native support for Microsoft's Cognitive Toolkit as well as Google's TensorFlow. Brainwave can speed up the predictions from machine learning models.

Apache Kafka gets SQL support

Apache Kafka, the key component in many data pipeline architectures, is getting SQL support. San Francisco-based Confluent has released an open source streaming SQL engine called KSQL that enables developers with continuous, interactive queries on Kafka.

The latest announcement is quite important for businesses that need to respond to SQL queries on Apache Kafka. The same functionality was earlier limited to Java or Python APIs.

Google releases Android 8.0 Oreo with new developer tweaks

Google has released Android 8.0 Oreo as the next iteration of its open source mobile platform. The latest update has a list of tweaks for developers to let them build an enhanced user experience.

"In Android 8.0 Oreo, we focused on creating fluid experiences that make Android even more powerful and easy to use," said Android's VP of engineering, Dave Burke in a blog post.

Android 8.0 is a result of months of testing by developers and early adopters who installed and tested its preview build on their devices. Also, it is designed to make the Android ecosystem more competitive with Apple's iOS.

Android 8.0 Oreo comes with the picture-in-picture mode that enables developers to provide an advanced multi-tasking experience on their apps. The feature was originally available on Android TV but is now on mobile devices, enabling users to simultaneously run two apps on the screen. Google has added a new object to enable the picture-in-picture mode. The object, called PictureInPictureParams, specifies properties such as the active app's preferred aspect ratio.

Android Oreo features other consistent notifications too. There are changes such as notification channels, dots and timeout. You just need to use a specific method to make notifications through your apps better on Android 8.0. Google has also added features such as downloadable fonts, and adaptive icons to upgrade the interface of existing apps. Likewise, the platform has WebView APIs and support for Java 8 language features. There are also the ICU4J Android Framework APIs that reduce the APK footprint of third-party apps by not compiling the ICU4J libraries in the app package.